



An-Najah National University

Faculty of Information Technology & Artificial Intelligence

Database Administration Course Homework

Data Warehouse Design Report: Movie Rental System

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Table of Contents

1. Introduction	3
2. Business Questions	3
3. Dimensional Model Design.....	5
3.1 Selected Business Processes	5
3.2 Fact Tables	5
3.3 Dimension Tables	6
4. Dimensional Model Diagram	7
5. ETL Design.....	7
5.1 Extract	7
5.2 Transform	8
5.3 Load	8
6. Data Quality Considerations.....	9
7. Conclusion	9

1. Introduction

The purpose of this assignment is to design a high-level Data Warehouse for a movie rental business, transitioning from a highly normalized Online Transaction Processing (OLTP) system to an Online Analytical Processing (OLAP) structure.

While the current OLTP system is optimized for fast, reliable daily transactions (inserting rentals, updating payments) and maintains data integrity through 3rd Normal Form (3NF), it is highly inefficient for historical reporting. Complex queries requiring multiple joins across tables (like customer, address, city, country, rental, payment) slow down the system.

The proposed Data Warehouse utilizes Dimensional Modeling (a Star Schema) to denormalize the data, organizing it into Facts (measurable events) and Dimensions (context), making it optimized for fast, intuitive analytical querying and business decision-making.

2. Business Questions

The data warehouse is designed to answer key analytical questions that drive strategic business decisions.

Revenue & Profitability:

-Which films and categories generate the highest revenue? (Helps in budget allocation for purchasing new inventory).

-How does revenue trend by month, quarter, and year? (Essential for financial forecasting and identifying seasonal trends).

Customer Behavior:

-Which customers generate the highest revenue, and which rent the most films? (Allows for targeted marketing and loyalty program rewards).

-What are the most active customer locations (cities/countries)? (Guides expansion strategies or localized marketing campaigns).

Operational Efficiency:

-Which stores generate the highest number of rentals and revenue? (Identifies high-performing vs. underperforming locations).

-Which staff members process the highest number of transactions? (Useful for employee performance reviews and staffing schedules).

-Which films are returned late most often, and what is the average rental duration? (Helps in adjusting inventory levels and late fee policies).

3. Dimensional Model Design

3.1 Selected Business Processes

The core business processes identified from the OLTP schema are:

- Movie Rentals: The act of a customer checking out a movie.
- Payments: The financial transaction associated with a rental.

3.2 Fact Tables

1. Fact_Rental

- Business Process: Film rental transactions.
- Grain: One row per individual movie rental transaction.
- Dimensions: Dim_Date (Rental Date, Return Date), Dim_Customer, Dim_Film, Dim_Store, Dim_Staff.
- Measures: Rental_Duration_Days (calculated), Expected_Duration_Days, Is_Late (1 or 0 flag), Late_Fee_Amount.

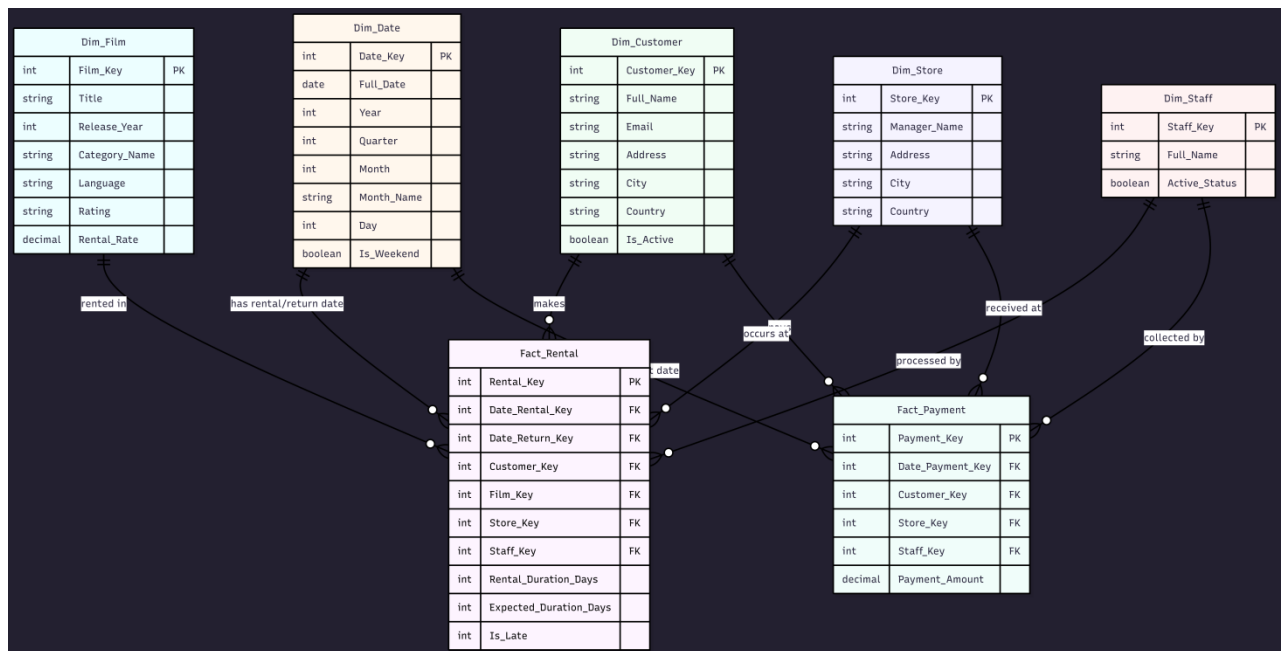
2. Fact_Payment

- Business Process: Payment processing.
- Grain: One row per payment transaction.
- Dimensions: Dim_Date (Payment Date), Dim_Customer, Dim_Store, Dim_Staff.
- Measures: Payment_Amount.

3.3 Dimension Tables

Dimension Name	Source OLTP Tables	Main Attributes	Purpose
Dim_Date	N/A (Generated)	Date_Key, Date, Day, Month, Quarter, Year, Is_Weekend	Allows slicing data by time and identifying trends. Serves as a role-playing dimension (Rental, Return, Payment dates).
Dim_Film	film, category, language, film_category	Film_Key, Title, Release_Year, Rating, Category_Name, Language, Rental_Rate	Centralizes all movie details. Category and Language are denormalized into this table.
Dim_Customer	customer, address, city, country	Customer_Key, Full_Name, Email, Address, City, Country, Is_Active	Stores customer demographics. Geographic data is rolled up into this dimension.
Dim_Store	store, address, city, country	Store_Key, Store_Manager, Address, City, Country	Tracks the physical locations where transactions occur.
Dim_Staff	staff	Staff_Key, Full_Name, Role, Email, Active_Status	Tracks the employees processing the transactions.

4. Dimensional Model Diagram



5. ETL Design

5.1 Extract

Data will be extracted from the source OLTP database. Initially, a full load will be required. Subsequent loads will be incremental based on timestamps (e.g., last_update columns).

Source Tables Needed: rental, payment (for Facts); customer, address, city, country, film, category, language, film_category, store, staff (for Dimensions).

5.2 Transform

Data will be cleansed, integrated, and transformed in a staging area:

- **Data Integration (Joins):** Join customer + address + city + country to create a flat Dim_Customer record. Join film + film_category + category to create a flat Dim_Film record.
- **Data Cleansing & Formatting:** Concatenate first_name and last_name into Full_Name. Standardize text cases for cities and countries.
- **Handling Missing Values:** For films that have not yet been returned, set the Return Date Key to a default future date (e.g., 99991231) or -1 to indicate "Unknown/Not Returned".
- **Deriving Measures:** Calculate Rental_Duration_Days by finding the date difference between rental_date and return_date. Compare this to the film's allowed rental period to flag Is_Late (1 or 0).
- **Surrogate Keys:** Replace operational IDs (Business Keys) with auto-incrementing integer Surrogate Keys (SKs) to isolate the warehouse from operational system changes.

5.3 Load

The loading phase must follow referential integrity rules (Dimensions first, Facts second):

- Load Dim_Date: Usually pre-populated with 10-20 years of calendar data.
- Load Dimensions (Dim_Film, Dim_Customer, Dim_Store, Dim_Staff):
- Insert new records and generate new Surrogate Keys.
- Handle updates using Slowly Changing Dimensions (SCD). For example, use SCD Type 1 (overwrite) for correcting a misspelled customer name, and SCD Type 2 (create a new row with active flags) if a customer moves to a new city, to preserve historical reporting.
- Load Fact Tables (Fact_Rental, Fact_Payment): Look up the Surrogate Keys from the newly loaded dimensions based on the OLTP Business Keys, then insert the transaction rows.

6. Data Quality Considerations

To ensure trust in the data warehouse, the following data quality rules must be checked during the ETL process:

- **Referential Integrity Check:** Every Foreign Key in a Fact table must match a valid Surrogate Key in the corresponding Dimension table (No orphaned facts).
- **Logical Date Validation:** Return_Date must always be greater than or equal to Rental_Date.
- **Null Checks:** Critical fields like Payment_Amount or Customer_Key cannot be NULL.
- **Deduplication:** Ensure no duplicate transaction IDs from the OLTP system are loaded into the Fact tables multiple times.

7. Conclusion

By restructuring the highly normalized OLTP movie rental schema into a dimensional Star Schema, the business is now equipped with a powerful analytical tool. The separation of facts (rentals and payments) from denormalized dimensions (customers, films, time, geography) drastically simplifies complex querying. This architecture supports rapid BI dashboarding, empowering management to easily track revenue trends, identify top-performing stores, and understand customer behaviors to drive strategic growth.